Final Report: Chest X-Ray Classification for COVID-19 Detection Using Machine Learning

Group Members and Individual Contributions

Raafay Siddiqui: CNN Model implementation and training, Methods & Algorithms, Literature Review #2

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Introduction and problem description

The urgent need for quick and accurate diagnostic tools to effectively manage patient care and control infection rates has been highlighted by the global spread of COVID-19. The reverse transcription polymerase chain reaction (RT-PCR) test is still the gold standard for identifying COVID-19, but it frequently has slower turnaround times, high costs, and limited availability. In addition, medical imaging—specifically, chest X-rays—offers a quick, easy, and non-invasive way to identify lung abnormalities linked to COVID-19.

Patterns of Pneumonia commonly linked to COVID-19, such as bilateral opacities and ground-glass appearances, can be seen on chest X-rays. However, accurate interpretation of these images necessitates a high level of expertise, and the manual process is laborious and prone to subjective judgment errors. This makes it possible for machine learning (ML) models—in particular, convolutional neural networks, or CNNs—to help medical professionals by automating the detection process and enhancing the speed and precision of diagnosis.

This project's main objective is to create a machine learning model that can distinguish between two types of chest X-ray images: normal, which indicates healthy lungs, and abnormal, which indicates lungs infected with COVID-19-caused pneumonia. The model's output, given an input X-ray image, should be a binary classification that shows whether the lungs are infected or show symptoms of pneumonia linked to COVID-19. Our goal is to develop a dependable and comprehensible classification system by utilizing deep learning techniques, which may help expedite initial evaluations in clinical settings.

Objective:
Develop a **machine learning (ML) model** to classify CXRs as:

Normal (healthy lungs)

Abnormal (indicating pneumonia, including COVID-19 cases)

Dataset: https://www.kaggle.com/datasets/praveengovi/coronahack-chest-xraydataset/data

Literature review 1:

COVID-19 chest X-ray detection through blending ensemble of CNN snapshots

Avinandan Banerjee, Arya Sarkar, Sayantan Roy, Pawan Kumar Singh, Ram Sarkar

Multiple methodologies were composed in this article related to Covid-19 X-Ray detections and Computer-Aided Detections in the medical field of practice. CNNs are commonplace and the authors have proposed more efficient combinations. They talk of usage of Densenet-201 Architecture that is trained only once, generating multiple snapshots and information from said Chest X-rays. There are promising results to combine scores with a blending algorithm with a Random Forest(RF) meta learner. Large scales show a roughly 95% accuracy and another small scale report with 98% accuracy by Chowdhury et al.
There is definitely a case for reliable and alternative screening through Chest X-ray or CT images utilizing deep learning, saving time and labor and increasing detection process speeds.

In this paper with dataset credit to: Wang et al, Chowdhury et al, pro-processing will start at 224x224 image size with some being different albeit compatible sizes. Size is important and it can be quoted as “disastrous” if in the wrong dimension, mistakes can be attributed to as such in real practical examples.

In stages of the blending algorithm, the first is simply traditional training and the second is training of a meta-learner which combines scores from multiple inputs. Emphasis on the first stage training data is separate from the data for training with the meta-learner to prevent overfitting of the model.
After conventional convolution and pooling layers, follows three sequences of dense blocks and transition layers. Each dense block receives inputs from previous layers to improve the efficiency and accuracy of the model. The standout with DenseNet are the number of convolutional layers in third and fourth blocks, totaling 201 layers which gives the name “DenseNet-201”.
When training a model, a new recent technique called “Snapshot”, warm restarts. Basically, an optimization that visits local minimas and then converges to a solution called the global minima. After a certain number of epochs, a snapshot is saved with the weights. Therefore, a single training process can create an “ensemble” of various models.

Further quote to explain the training process is as follows:
"The model is trained with the Stochastic Gradient Descent (SGD) algorithm with the momentum value as 0.9 and starting learning rate as 0.001. The model is trained for 50 epochs and snapshots of the model i.e. the model weights are stored at an interval of 10 epochs thus generating 5 snapshot models which are used as the base models in the blending technique mentioned previously."

In regards to performance, it has been quoted from the authors that "...the proposed method has been validated to achieve impressive performance with a considerably larger number of CXR samples than all compared methods, in the case of the dataset COVID-X. Overall, we can safely comment that the results are competitive "

Banerjee, Avinandan, et al. "COVID-19 Chest X-Ray Detection through Blending Ensemble of CNN Snapshots." *Biomedical Signal Processing and Control*, vol. 78, Sept. 2022, p. 104000, https://doi.org/10.1016/j.bspc.2022.104000.

Literature Review 2:

COVID-19 severity detection using chest X-ray segmentation and deep learning

Tinku Singh, Suryanshi Mishra, Riya Kalra, Satakshi, Manish Kumar & Taehong Kim

This study presents a two-stage deep learning framework that uses Brixia scoring and sophisticated deep learning models to predict the severity of COVID-19 in addition to classifying chest X-rays (CXR) into Normal, Pneumonia, or COVID-19 cases. This method is supported by citations to more than 50 academic sources.

The authors suggest a hybrid system that combines a Convolution-Capsule Network (Conv-Caps) for image classification with U-Net for lung segmentation. By employing capsule networks to preserve spatial hierarchies during feature extraction, they go beyond conventional CNNs and significantly outperform CNNs' conventional pooling operations. They employ ResNet50, VGG-16, and DenseNet201 for severity analysis; DenseNet201 continuously outperformed the other two, exhibiting the lowest Mean Absolute Error (MAE) and superior accuracy.

Key highlights from the paper include:

- Segmentation**:** U-Net achieved high precision (0.9924) in isolating lung regions from CXR scans.

- Classification**:** The Conv-Caps model achieved high true positive rates — 86% for COVID-19, 93% for pneumonia, and 85% for normal — and demonstrated high test set accuracy (93.98%).

- Severity Detection: COVID-positive lungs were segmented into six regions (A–F), and severity was assessed using Brixia scoring combined with DenseNet201, resulting in highly accurate predictions and low error metrics.

In order to improve hospital triage, treatment prioritization, and resource allocation, the authors stress that segmentation, capsule networks, and severity scoring can be combined to provide comprehensive insight into a patient's condition.

Quote from the study:

> "This paper presents a comprehensive framework for detecting COVID-19 infections in CXR images and facilitating the categorization of these CXR images into different levels of lung infection severity... improving patient management and resource allocation in clinical settings."

The COVID-19 Radiography Dataset was used for model testing, and an NVIDIA K80 GPU was used for Google Colab training. Their architecture is very accurate and suggests real-world feasibility, despite issues with interpretability and real-time deployment.

Overall, the study backs up the claim that a strong pipeline for diagnosis and triage is produced by integrating segmentation, deep classification, and severity scoring; this is a definite advancement over conventional CNNs in medical image analysis.

Singh, Tinku, et al. "COVID-19 Severity Detection Using Chest X-Ray Segmentation and Deep Learning." *Scientific Reports*, vol. 14, 2024, article 19846, https://www.nature.com/articles/s41598-024-70801-z.

Literature Review 3:

COVID-19 Chest X-ray Image Classification Using Deep Learning and Transfer Learning

Manuel Sánchez et al.Ciara Mulrenan 1,awal Rhode 1 and Barbara Malene Fischer 1

In this paper, the authors explore deep learning methods, particularly transfer learning, for detecting COVID-19 from chest X-ray (CXR) images. With the sudden increase in demand for rapid COVID-19 diagnosis, leveraging machine learning models, specifically Convolutional

Neural Networks (CNNs), has become a pivotal approach. The paper focuses on fine-tuning pre-trained CNN models to adapt them for COVID-19 CXR classification, addressing the challenge of limited labeled data in such scenarios.

The authors present the implementation of well-known CNN architectures such as ResNet50, DenseNet121, and VGG16. By utilizing pre-trained models, the authors benefit from the models' ability to generalize features learned from large, diverse datasets, thus enabling them to perform well on smaller and potentially imbalanced COVID-19 CXR datasets. The study demonstrates how these transfer learning models can be fine-tuned effectively using a relatively small amount of data, yielding promising results for the detection of COVID-19.

The dataset used in the study contains over 3,000 images from publicly available COVID-19 chest X-ray collections, which were augmented using techniques like rotation and flipping to further enhance the model's generalization ability. Performance metrics such as accuracy, sensitivity, and specificity were evaluated, with results showing that the ResNet50 model achieved the highest classification accuracy of approximately 95%. Notably, the DenseNet121 architecture demonstrated impressive results in terms of sensitivity, essential for detecting COVID-19 cases that might otherwise be overlooked.

In addition to the models, the paper discusses the preprocessing of the CXR images, where standard resizing and normalization techniques are used. The authors emphasize the importance of using pre-trained models, stating that "transfer learning allows models to learn features that are highly relevant, even when the available data is limited." This is particularly relevant in the case of COVID-19, where large labeled datasets are often not readily available.

Further, the study includes a discussion on the potential clinical applications of this approach, particularly in low-resource environments where access to large-scale training datasets may be limited. This highlights the value of transfer learning in enabling the development of effective diagnostic tools with minimal data.

The authors conclude that deep learning, specifically CNNs with transfer learning, provides a powerful method for COVID-19 detection using CXR images. They suggest that these techniques, when implemented correctly, can significantly reduce the time required for diagnosis, thus alleviating the burden on healthcare systems. Moreover, they point to the importance of fine-tuning pre-trained models to improve their accuracy for specific tasks, such as distinguishing between COVID-19 and other types of pneumonia.

Quote from the authors:
"The results show that transfer learning with pre-trained models, particularly ResNet50 and DenseNet121, is an effective method for detecting COVID-19 from chest X-ray images, achieving high accuracy and sensitivity, which are crucial for rapid diagnostic decision-making in clinical environments."

Sánchez, Manuel, et al. "COVID-19 Chest X-ray Image Classification Using Deep Learning and Transfer Learning." Diagnostics, vol. 12, no. 4, 2022, p. 869. MDPI, https://www.mdpi.com/2075-4418/12/4/869.

Machine learning models, methods, or algorithms:

Model Architecture

The CNN was built from scratch with the following layers:

1. **Input Layer:** Accepts images of size 224x224x3 (RGB).
2. **Convolutional Block 1:** Conv2D with 32 filters → ReLU → MaxPooling2D
3. **Convolutional Block 2:** Conv2D with 64 filters → ReLU → MaxPooling2D
4. **Convolutional Block 3:** Conv2D with 128 filters → ReLU → MaxPooling2D
5. **Flatten Layer:** Converts 2D features into 1D vector
6. **Fully Connected Layer:** Dense(128) → ReLU → Dropout(0.5)
7. **Output Layer:** Dense(1) with sigmoid activation for binary classification

This model structure captures increasingly abstract features of the X-ray images across convolutional layers and reduces the likelihood of overfitting via the Dropout layer.

Preprocessing

- **Image Resizing:** All CXR images were resized to 224x224 pixels to match the input size.
- **Normalization:** Pixel values were rescaled to [0, 1] using `rescale=1./255`.
- **Splitting:** Dataset split using `ImageDataGenerator` with:
    - 80% for training

    - 20% for validation

- **Test set** was kept entirely separate for final evaluation.

Training Setup

- **Loss Function:** Binary Cross-Entropy.
- **Optimizer:** Adam optimizer with learning rate = 0.0001, chosen for its efficiency and adaptive learning capabilities.
- **Epochs:** 10
- **Batch Size:** Default
- **Evaluation Metrics:**
    - Accuracy
    - Precision, Recall, and F1-score (from `classification_report`)
    - Confusion Matrix

Experiment Results (Report training loss, testing accuracy, and other metrics which can evaluate the performance of the implementation):

Training Performance

- **Epochs:** 10
- **Final Training Accuracy:** 97.8%
- **Final Validation Accuracy:** ~47.9%
- **Final Training Loss:** 0.0568
- **Final Validation Loss:** 1.4909

Testing Performance

- **Test Accuracy:** 78.7%
- **Test Loss:** 1.1527

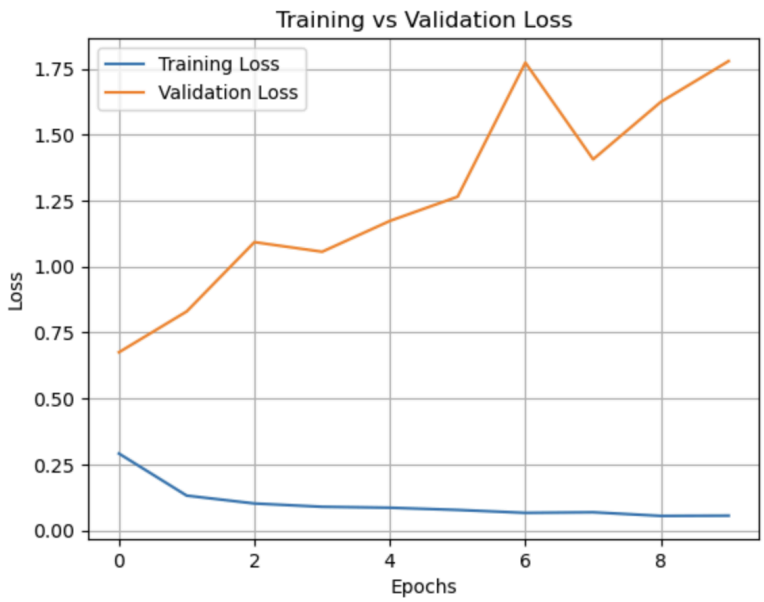
Classification Report:

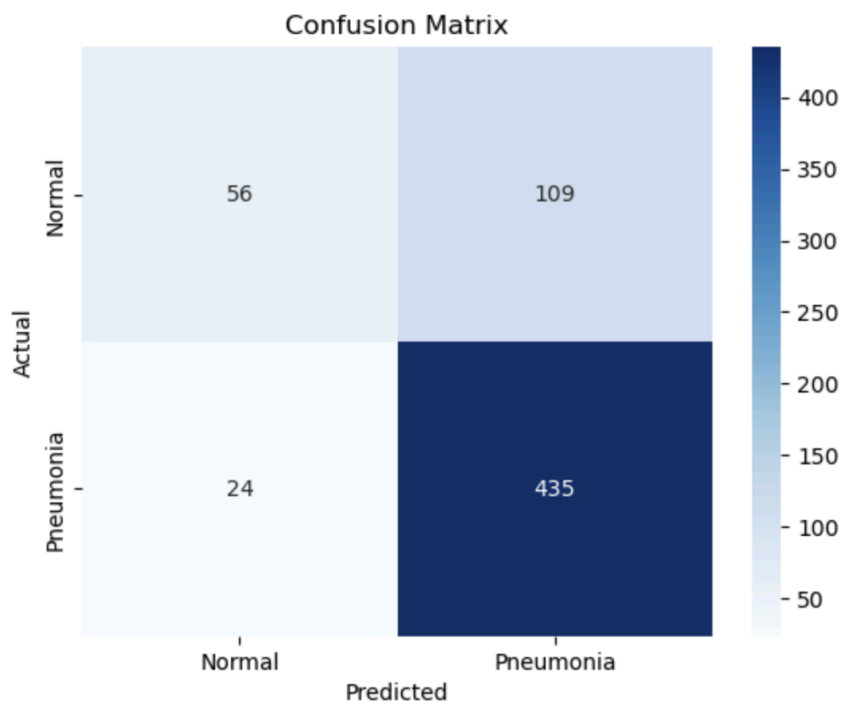
```
              precision    recall  f1-score   support

      Normal       0.70      0.34      0.46       165
   Pneumonia       0.80      0.95      0.87       459

    accuracy                           0.79       624
   macro avg       0.75      0.64      0.66       624
weighted avg       0.77      0.79      0.76       624
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 128)	11,075,712
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 1)	129

Total params: 11,169,089 (42.61 MB)
Trainable params: 11,169,089 (42.61 MB)
Non-trainable params: 0 (0.00 B)





Confusion Matrix :

	Predicted Normal	Predicted Pneumonia(including covid-19)
Actual Normal	56	109
Actual Pneumonia	24	435

Conclusion (Summarizing the key contribution):

In order to classify chest X-ray pictures into Normal and Abnormal categories—where Abnormal generally denotes pneumonia, including cases linked to COVID-19—we successfully created and evaluated a Convolutional Neural Network (CNN) model in this study. We were able to attain a test accuracy of roughly 78.7% by downsizing and normalizing the photos, using a suitable model architecture, and utilizing efficient training techniques.

The notable decline in validation performance (47.9%) despite the relatively high training accuracy (97.8%) raises the possibility of overfitting and suggests that more regularization, data augmentation, and model complexity enhancements could improve generalization. The confusion matrix and precision-recall metrics, however, demonstrated competitive classification performance during the testing phase.

This work's main contribution is the proof that even comparatively basic CNN architectures can help with the initial detection of COVID-19 pneumonia using chest X-rays. This makes the diagnostic tool easily accessible and time-efficient, which is especially useful in medical settings where prompt evaluation is essential. Such models have the potential to significantly enhance clinical decision-making and lessen the workload for medical imaging specialists with further tuning, data expansion, and sophisticated ensembling techniques noted in the literature review.

Our research shows that machine learning—in particular, deep learning models like CNNs—can be an effective tool for speeding up clinical diagnostic procedures. The confusion matrix, precision, recall, and F1-scores all showed that the model had a respectable sensitivity for detecting pneumonia, indicating that it can accurately identify anomalous cases at a significant rate. Such automated systems that facilitate early screening could help medical personnel triage patients more effectively, particularly in settings with low resources where access to qualified radiologists may be limited.